CASE 15
CHEDIAK–HIGASHI SYNDROME

A defect of intracellular vesicle trafficking.

Cytoplasmic vesicles are essential components of all cells in the human body. These small membrane-enclosed sacs participate in diverse cellular functions, including transport within the cell and with the exterior, storage of nutrients, digestion of cellular waste and foreign products, and processing of proteins. They also serve as closed chambers for multiple chemical reactions that could otherwise damage the cell if they were to occur outside the vesicle. The types of vesicles and their specific functions vary between different cells, although most of them would appear as undistinguishable intracellular granules under the microscope.

Correct trafficking of vesicles is as important as the function of the vesicle itself. The process through which cells absorb molecules or structures from the exterior by invaginating the cell wall and forming a vesicle is called endocytosis. In contrast, release of the contents of intracellular vesicles to the exterior is called exocytosis. Vesicles and their trafficking in the cell are particularly important for normal functioning of the immune system. They are essential in the first line of defense, in which cells of the innate immune system, such as neutrophils and macrophages, engulf invading microorganisms through a form of endocytosis called phagocytosis (Fig. 15.1). This newly created vesicle is called a phagosome. The phagosome then fuses with a lysosome, a different type of vesicle that contains the machinery to kill the invading organism and digest its components. This fused vesicle, called

TOPICS BEARING ON THIS CASE:

Phagocytosis

Endocytosis and processing of antigen

Killing by cytotoxic T cells and NK cells

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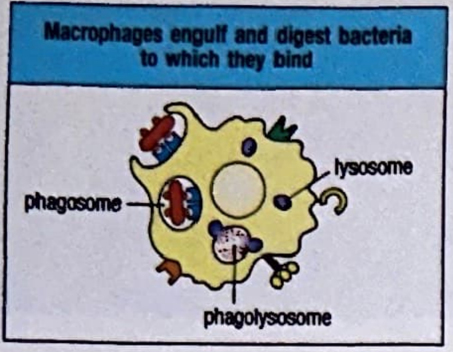


Fig. 15.1 **Cell vesicles involved in phagocytosis.** A phagocyte, such as a macrophage, binds to bacteria and engulfs them, creating a phagosome. The phagosome fuses with a lysosome, which contains hydrolytic enzymes, creating a vesicle called the phagolysosome, in which the bacteria are digested and killed.

a phagolysosome, degrades its contents using several mechanisms, including the production of reactive oxygen species such as superoxide, and the action of both proteolytic enzymes, such as lysozyme, and defensins and other antimicrobial peptides.

Other immune cells that engage in endocytosis are antigen-presenting cells such as dendritic cells. In this case, the cell captures a foreign macromolecule and degrades it, but the main function in this instance is the loading of peptide onto antigen-presenting proteins (MHC proteins), which will expose the newly acquired antigen to examination by T cells. T cells that carry receptors recognizing the foreign antigen in association with MHC proteins get activated and trigger the adaptive immune response.

Secretory vesicles, whose function is to release their contents to the extracellular space or into other cells through exocytosis, also have an essential function in the immune response. Cytotoxic T lymphocytes (CD8 T cells) and natural killer cells (NK cells) recognize other host cells that have been infected by viruses or intracellular bacteria and kill them by transporting secretory vesicles to the cell surface, where they fuse with the cell membrane and release their contents of enzymes and other mediators, such as perforin and granzymes, thus inducing apoptosis of the target cell.

In addition to their role in the functioning of the immune system, vesicles have an important role in pigmentation of the skin and eye, synapses between neurons, and platelet activation. As we shall illustrate, a defect in vesicle trafficking can lead to a devastating disease affecting many different functions of the immune system, as well as other organs and systems.

The case of Shweta Amdra Devi: the cause of recurrent infections is found in a peripheral blood smear.

Shweta was seen by the immunologists at the Children's Hospital at the age of 4 years, with a history of recurrent ear infections and two episodes of pneumonia. At 2 years of age she had also had cellulitis (bacterial infection of the skin) and lymphadenitis (lymph node infection) due to *Staphylococcus aureus*, which required intravenous antibiotics. In addition, Shweta had a history of bruising easily and multiple episodes of mild nosebleeds.

Shweta was born at full term and with a normal birth weight. Her umbilical cord separated normally at 10 days. Her parents reported that ever since Shweta was a baby, she seemed to be bothered by bright lights. She also had fair skin and hair, which was much lighter than the skin and hair of her parents and siblings and was unusual for her Indian descent. This was the reason she had been named Shweta, which means "fair complexioned" in the Hindi language.

Recurrent infections. Fairer complexion and lighter hair color than other family members.

Her parents were second cousins, who had recently emigrated to the United States from a small town near Bangalore, India. Shweta had two healthy siblings, a 10-year-old boy and a 17-year-old girl. There were no family members with a history of recurrent infections.

On physical examination, Shweta was relatively thin and small for her age. She had a mild fever of 37.7°C and slightly fast breathing. Her skin was fair and she had light brown hair with a silvery sheen. She had an atrophic scar on her left leg, the consequence of an old skin infection. The irises of her eyes were gray. An ear examination revealed a perforation and pus drainage in the right tympanic membrane, and she had moderate cervical lymphadenopathy (enlarged lymph nodes). Examination of

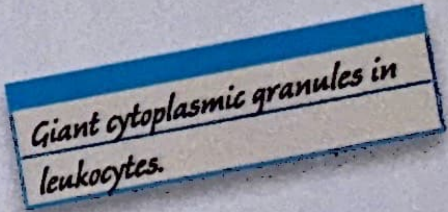
her lungs revealed very poor aeration and crackles in the right lower lobe. Her abdomen had mild hepatosplenomegaly (enlarged liver and spleen).

A chest radiograph confirmed that Shweta had a consolidation in the right lower lobe and a moderate pleural effusion, thus confirming a new episode of pneumonia. Aspiration of the pleural fluid grew *S. aureus*, which was broadly sensitive to antibiotics. She was admitted to the hospital and started on intravenous nafcillin (an antistaphylococcal antibiotic), after which she slowly recovered.

Laboratory testing revealed a white blood cell count of 5200 μl^{-1} (normal), of which 21% (1100 cells μl^{-1}) were neutrophils (mildly decreased), 68% were lymphocytes (normal), and 8% were monocytes (normal). Serum immunoglobulins and complement titers were normal. Because of her history of infections and current staphylococcal pneumonia, neutrophil function was measured with a nitro blue tetrazolium (NBT) test, which assays the capacity of the NADPH oxidase in neutrophils to produce the superoxide that reduces the NBT dye from yellow to blue (see Case 26). The results showed that Shweta's neutrophils could reduce NBT normally. The laboratory technician examining the peripheral blood smear observed giant cytoplasmic granules in her leukocytes, which stained positive with myeloperoxidase. She called the physicians in charge of Shweta's care and told them, "this patient has Chediak–Higashi syndrome."

After Shweta recovered from the acute infection, she was started on prophylactic antibiotics, and plans were made for a bone marrow transplant. Shweta's brother proved to be a full HLA match and donated the bone marrow. Shweta was successfully transplanted with full resolution of infections. Years later, during adolescence, Shweta developed a progressive neurological disease characterized by weakness, tremors, and ataxia, which confined her to a wheelchair.

Fig. 15.2 A child with Chediak–Higashi syndrome (CHS). Note the white hair (albino) and the silvery sheen of the hair.



Chediak–Higashi syndrome.

Patients such as Shweta have a very rare and severe disease called Chediak–Higashi Syndrome (CHS). This is an autosomal recessive disorder clinically characterized by recurrent bacterial infections and partial absence of pigmentation of skin, hair, and eyes (oculocutaneous albinism) (Fig. 15.2). Affected patients also have a tendency to bleeding, due to platelet dysfunction, which is usually mild to moderate. If patients survive into adolescence or early adulthood, most will develop progressive neurological defects, including cerebellar ataxia, central nervous system atrophy, seizures, peripheral neuropathy, and cognitive defects. In addition, most patients with CHS undergo at some point an accelerated phase of uncontrolled lymphocyte proliferation and lymphohistiocytic infiltration characterized by fever, lymphadenopathy, hepatosplenomegaly, and pancytopenia (reduction of platelets and of white and red blood cells), which is usually lethal.

The diagnosis of CHS can be made by examination of a peripheral blood smear for the distinctive giant cytoplasmic granules (vesicles) in leukocytes and platelets (Fig. 15.3). A similar phenotype of giant intracellular granules was known in the *beige* mouse strain, which has hypopigmentation of the coat (Fig. 15.4). The *beige* mouse was crucial in the identification of the human gene responsible for CHS, which was named *CHS1* (*Chediak Higashi Syndrome 1*). This gene, which is also

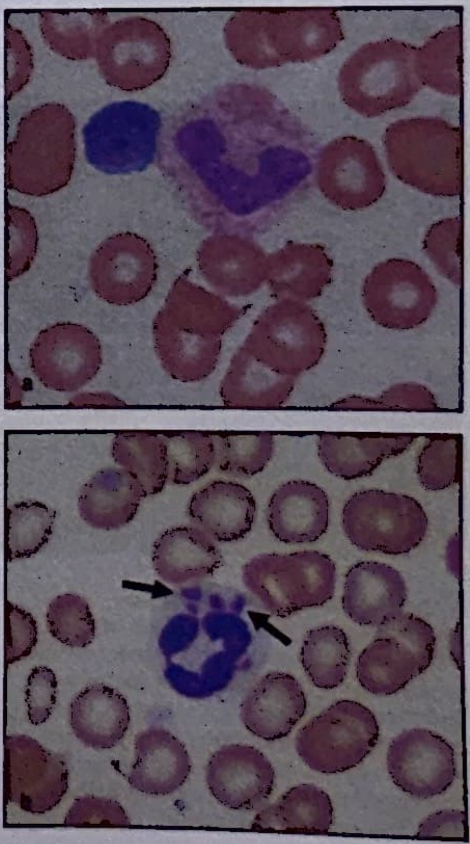


Fig. 15.3 Large intracellular granules in leukocytes are typical of CHS. Compare the peripheral blood smear from a healthy person (panel a) with that from a patient with Chediak–Higashi syndrome (panel b). The large cell in the center of both photographs is a neutrophil, distinguishable by the irregular lobed nucleus (stained purple). Note the abnormally large granules in the cytoplasm of the neutrophil from the patient with CHS. Preparations stained with hematoxylin and eosin.

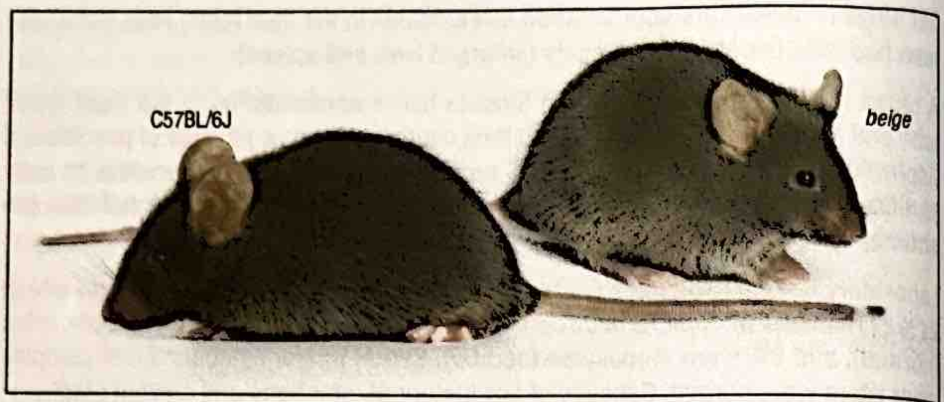


Fig. 15.4 The beige mouse is an animal model for CHS. The color of the fur of the *beige* mouse (on the right) is lighter than the C57BL/6J wild-type mouse because of abnormalities in pigmentation associated with mutations in the *CHS1* gene.

known as *LYST* (*LYSosomal Trafficking regulator*), is part of the beige and Chediak-Higashi syndrome (BEACH)-domain containing family of proteins involved in vesicle formation and trafficking. In addition to humans and mice, other mammals with mutations in *CHS1* have been identified, including the Aleutian mink, whose coat has a blue tinge.

All cells from patients with CHS have an abnormal clustering of giant lysosome-like vesicles around the nucleus. It is not yet clear how mutations in *CHS1/LYST* cause the vesicle defect, because the precise biological function of the protein is unknown. It is suspected that abnormal organellar protein trafficking may lead to aberrant fusion of vesicles and a failure to transport lysosomes to the appropriate location in the cell. Most mutations in *CHS1* are nonsense or null mutations that lead to an absence of the protein, although some cases with milder phenotypes due to missense mutations have been described.

The main immunological defect of patients with CHS is in innate immunity, affecting the first line of defense to pyogenic infections of the skin and respiratory tract. Infections are frequent and usually severe, beginning shortly after birth. Commonly implicated microorganisms include the bacteria *Staphylococcus aureus*, *Streptococcus pyogenes*, and *Streptococcus pneumoniae*, and occasionally fungi such as *Candida* or *Aspergillus*. Neutrophil counts are mildly to moderately decreased, probably as a result of the destruction of neutrophils in the bone marrow. These cells also have decreased intracellular microbicidal activity. Affected neutrophils and monocytes have a chemotactic and migratory capacity that is about 40% that of normal cells. This may be due in part to the fact that the large fused granules impair the cells' ability to move. In addition, cytotoxic T lymphocytes and NK cells have severely impaired cytotoxicity because of their inability to secrete the granules that contain lytic proteins such as granzymes and perforin. Studies of B cells of affected patients also suggest that these are less able to load peptide onto MHC class II molecules and have decreased antigen presentation compared with normal B cells.

Fig. 15.5 Abnormalities of hair pigmentation are diagnostic of CHS. Photomicrograph of a hair shaft from an unaffected individual (upper panel) and a patient with CHS (lower panel). Note the uniform distribution of the pigment melanin in the normal hair shaft and the abnormal speckles of pigment in the hair from the patient with CHS.

The pigmentation defect of CHS is due to the inability of melanocytes, the cells that produce the pigment, to transfer pigment-containing secretory granules (melanosomes) to keratinocytes and other epithelial cells (Fig. 15.5). Platelet dysfunction is due to a reduction in a particular group of vesicles called platelet dense bodies that participate in sustaining platelet aggregation. The pathogenesis of the neurological disease that affects patients with CHS is believed to be due to problems of vesicular trafficking in neurons and glia.

CHS is treated with prophylactic antibiotics and aggressive management of infections. Bone marrow transplantation corrects the immunological and hematological defects, and prevents the accelerated phase of CHS. However, manifestations of CHS in nonhematopoietic organs, particularly the oculocutaneous albinism and the progressive neurological disease, are not modified by bone marrow transplantation.

Questions.

1 Why do you think Shweta's NBT test was normal?

2 Besides the observation of giant granules in the leukocytes of a peripheral blood smear, what other non-invasive test can be helpful to confirm the diagnosis of CHS?

3 The accelerated phase of CHS consists of massive organ infiltration by lymphocytes. Which of the immunological defects in this disease would you think is involved?

4 Why does bone marrow transplantation not affect the oculocutaneous albinism and neurological disease?

5 Do you know of another immunodeficiency caused by mutations in a BEACH protein family member?